Trig Ceva and the ratio lemma

L4-X1 Level 4 Olympiad · Geometry · Beyond the textbook

Kevin Hu
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Trigonometric Ceva with proof, the ratio lemma and its uniqueness companion, and isogonal lines, for concurrence and for angle problems that angle chasing cannot finish.
By the end you can turn angles at a vertex into a ratio on the opposite side, and back, with the ratio lemma; prove concurrence with trigonometric Ceva and find unknown angles with it, including the uniqueness step; recognise isogonal lines and use Steiner's ratio and isogonal conjugates.
Prerequisites: Olympiad angle chasing (L4-GE1), Trigonometry and coordinates (L3-GE2). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Prove that the three altitudes of an acute triangle ABC are concurrent. Let the altitudes be AD, BE, CF. In the right-angled triangle ABD we have $\angle BAD = 90^\circ - B$, and in the same way $\angle DAC = 90^\circ - C$, $\angle CBE = 90^\circ - C$, $\angle EBA = 90^\circ - A$, $\angle ACF = 90^\circ - A$ and $\angle FCB = 90^\circ - B$. Hence

$$\frac{\sin\angle BAD}{\sin\angle DAC}\cdot\frac{\sin\angle CBE}{\sin\angle EBA}\cdot\frac{\sin\angle ACF}{\sin\angle FCB} = \frac{\cos B}{\cos C}\cdot\frac{\cos C}{\cos A}\cdot\frac{\cos A}{\cos B} = 1,$$

and by trigonometric Ceva (Theorem 3 below) the altitudes concur. No length on BC was computed.

Ceva's theorem (Giovanni Ceva, 1678) is taught in school with the lengths BD, DC, CE, EA, AF, FB. Olympiad problems usually give angles at the vertices instead, and then these lengths are unknown. The sine form of Ceva, and the lemma behind it, which olympiad trainers call the *ratio lemma*, convert angles at a vertex into a ratio on the opposite side and back. Both are folklore in olympiad training; L4-GE2 states them, and here we prove them, prove the uniqueness statement that most solutions need and most textbooks omit, and use them on isogonal lines. The sine rule and area formula (L3-GE2) and angle chasing (L4-GE1) are assumed.

1.1 1. The ratio lemma

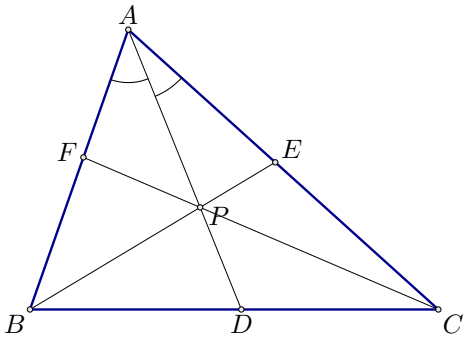
Theorem 1 (ratio lemma).
Let D be a point of line BC other than B and C. Then

$$\frac{BD}{DC} = \frac{AB\sin\angle BAD}{AC\sin\angle DAC}.$$

Proof. The sine rule in triangles ABD and ADC gives $\dfrac{BD}{\sin\angle BAD} = \dfrac{AB}{\sin\angle ADB}$ and $\dfrac{DC}{\sin\angle DAC} = \dfrac{AC}{\sin\angle ADC}$. The angles $\angle ADB$ and $\angle ADC$ are supplementary if D lies between B and C, and equal otherwise; in both cases their sines are equal. Dividing the two equations gives the result. ■

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Two special cases are worth remembering. If AD bisects angle A, we get the angle bisector theorem $BD/DC = AB/AC$. If M is the midpoint of BC, we get $AB\sin\angle BAM = AC\sin\angle MAC$: a median is a statement about angles at A.



Lemma 2 (uniqueness).
Let $0^\circ < \theta < 180^\circ$. On the interval $0^\circ < x < \theta$ the function $f(x) = \dfrac{\sin x}{\sin(\theta - x)}$ is increasing. Hence if $f(x) = f(y)$ with x, y in this interval, then $x = y$.

Proof. Expanding $\sin(\theta - x) = \sin\theta\cos x - \cos\theta\sin x$ gives $\dfrac{1}{f(x)} = \sin\theta\cot x - \cos\theta$. Since $\sin\theta > 0$ and $\cot x$ decreases on $(0^\circ, 180^\circ)$, the positive quantity $1/f(x)$ decreases as x increases. ■

Lemma 2 is the step most scripts omit. Trig Ceva gives an equation for an unknown angle; Lemma 2 says the equation has one solution, so a value that satisfies it is the answer.

1.2 2. Trigonometric Ceva

Theorem 3 (trigonometric Ceva).
Let D, E, F lie inside the sides BC, CA, AB of triangle ABC. Then AD, BE, CF are concurrent if and only if

$$\frac{\sin\angle BAD}{\sin\angle DAC}\cdot\frac{\sin\angle CBE}{\sin\angle EBA}\cdot\frac{\sin\angle ACF}{\sin\angle FCB} = 1.$$

Proof. By Theorem 1,

$$\frac{BD}{DC} = \frac{AB\sin\angle BAD}{AC\sin\angle DAC'}, \quad \frac{CE}{EA} = \frac{BC\sin\angle CBE}{BA\sin\angle EBA'}, \quad \frac{AF}{FB} = \frac{CA\sin\angle ACF}{CB\sin\angle FCB}.$$

Multiplying, the side lengths cancel, so the product of sines equals $\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}$, and Ceva's theorem gives both directions. ■

For a point P inside the triangle we write the condition as $\dfrac{\sin\angle PAB}{\sin\angle PAC}\cdot\dfrac{\sin\angle PBC}{\sin\angle PBA}\cdot\dfrac{\sin\angle PCA}{\sin\angle PCB} = 1$. Read it by going round the triangle: at each vertex the angle in the numerator is the one next to the side you reach first going $A \to B \to C \to A$.

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Corollary 4 (Ceva on a circle).
Let A_1, B_1, C_1 lie on the circumcircle of ABC, on the arcs BC, CA, AB not containing A, B, C. Then AA_1, BB_1, CC_1 are concurrent if and only if $A_1B \cdot B_1C \cdot C_1A = A_1C \cdot B_1A \cdot C_1B$.

Proof. The arc condition says that AA_1 meets side BC at an interior point, and so on. By the extended sine rule $A_1B = 2R\sin\angle BAA_1$ and $A_1C = 2R\sin\angle A_1AC$, so the condition is Theorem 3. ■

Example 1.
A point P inside triangle ABC satisfies $\angle PBC = 18^\circ$, $\angle PBA = 30^\circ$, $\angle PCB = 24^\circ$ and $\angle PCA = 54^\circ$. Find $\angle PAB$.

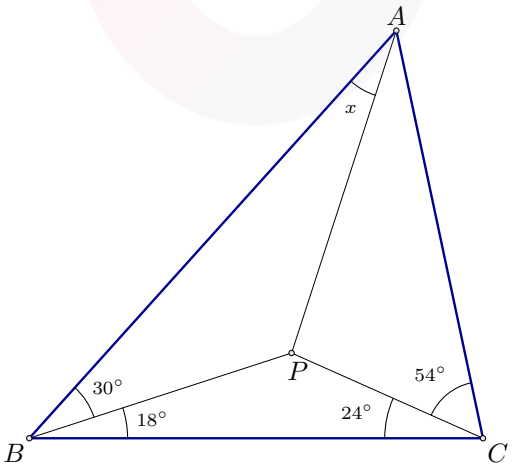
Solution.
We have $B = 48^\circ$, $C = 78^\circ$, so $A = 54^\circ$. Let $x = \angle PAB$, so $\angle PAC = 54^\circ - x$. Angle chasing first: $\angle BPC = 138^\circ$, $\angle APB = 150^\circ - x$ and $\angle APC = 72^\circ + x$. These add up to 360° for every x, so angle chasing produces no equation for x. A synthetic solution needs an auxiliary point, and there is no obvious one.
Trig Ceva gives an equation at once:

$$\frac{\sin x}{\sin(54^\circ - x)} = \frac{\sin 30^\circ \sin 24^\circ}{\sin 18^\circ \sin 54^\circ}.$$

A calculator gives about 0.813 for the right side, and $x = 24^\circ$ fits; we prove it. Since $\sin 54^\circ = \cos 36^\circ$,

$$\sin 18^\circ \cos 36^\circ = \frac{2\sin 18^\circ \cos 18^\circ \cos 36^\circ}{2\cos 18^\circ} = \frac{\sin 36^\circ \cos 36^\circ}{2\cos 18^\circ} = \frac{\sin 72^\circ}{4\cos 18^\circ} = \frac{1}{4}.$$

So the right side is $\frac{1}{2}\sin 24^\circ \cdot 4 = 2\sin 24^\circ = \dfrac{\sin 24^\circ}{\sin 30^\circ}$, which is $f(24^\circ)$ with $\theta = 54^\circ$. By Lemma 2, $x = 24^\circ$ is the only solution: $\angle PAB = 24^\circ$ and $\angle PAC = 30^\circ$.



Example 2.
The incircle of triangle ABC touches BC, CA, AB at D, E, F. Let X, Y, Z be the feet of the perpendiculars from D to EF, from E to FD and from F to DE. Prove that AX, BY, CZ are concurrent.

Solution.

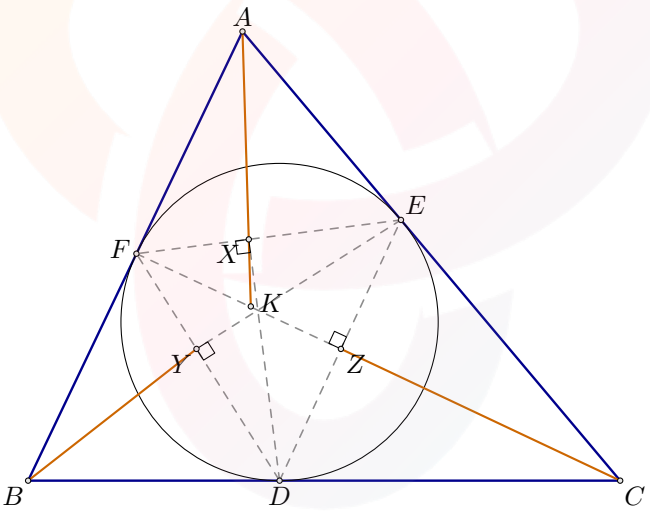
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Length Ceva would need the points where AX meets BC, which are hard to find. The angles at A are accessible through triangle AEF, in which AX is a cevian, so we use the ratio lemma there.

The angles of triangle DEF are $\angle EDF = 90^\circ - \frac{A}{2}$, $\angle DEF = 90^\circ - \frac{B}{2}$, $\angle DFE = 90^\circ - \frac{C}{2}$ (tangent-chord angle; for instance $\angle DEF = \angle BDF = 90^\circ - \frac{B}{2}$ as $BD = BF$). All are acute, so X lies inside segment EF and X is inside triangle AEF. The incircle is the circumcircle of DEF, so with r its radius, $DE = 2r\cos\frac{C}{2}$ and $DF = 2r\cos\frac{B}{2}$. Then

$$\frac{EX}{XF} = \frac{DE\cos\angle DEF}{DF\cos\angle DFE} = \frac{\cos\frac{C}{2}\sin\frac{B}{2}}{\cos\frac{B}{2}\sin\frac{C}{2}} = \frac{\tan\frac{B}{2}}{\tan\frac{C}{2}}.$$

In triangle AEF we have $AE = AF$, so the ratio lemma gives $\frac{EX}{XF} = \frac{\sin\angle EAX}{\sin\angle XAF} = \frac{\sin\angle XAC}{\sin\angle BAX}$. Hence $\frac{\sin\angle BAX}{\sin\angle XAC} = \frac{\tan\frac{C}{2}}{\tan\frac{B}{2}}$, and in the same way $\frac{\sin\angle CBY}{\sin\angle YBA} = \frac{\tan\frac{A}{2}}{\tan\frac{C}{2}}$ and $\frac{\sin\angle ACZ}{\sin\angle ZCB} = \frac{\tan\frac{B}{2}}{\tan\frac{A}{2}}$. The product is 1. Each line AX, BY, CZ meets the opposite side at an interior point, since X is inside the triangle, so Theorem 3 shows that they are concurrent.



1.3 3. Isogonal lines

Two lines through A are *isogonal* if each is the reflection of the other in the bisector of angle A; for lines meeting segment BC at D and E this says $\angle BAD = \angle EAC$. The reflection of the median from A is the A-symmedian. The altitude AH and the line AO are isogonal: $\angle BAH = 90^\circ - B$, and triangle AOC is isosceles with $\angle AOC = 2B$, so $\angle OAC = 90^\circ - B$ (for an acute triangle; directed angles handle the rest).

Theorem 5 (Steiner).
If D and E lie on segment BC and AD, AE are isogonal, then

$$\frac{BD}{DC}\cdot\frac{BE}{EC} = \frac{AB^2}{AC^2}.$$

Theorem 6 (isogonal conjugate). If P is inside triangle ABC, the reflections of AP, BP, CP in the bisectors of the angles at A, B, C meet at one point P^*.

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Proof. s. For Theorem 5, apply Theorem 1 to D and to E and multiply; since $\angle BAD = \angle EAC$ and $\angle DAC = \angle BAE$, the sines cancel and AB^2/AC^2 remains. For Theorem 6, reflecting the three lines swaps the two angles at each vertex, so each fraction in Theorem 3 is replaced by its reciprocal. The product is still 1 and the reflected lines are still inside the angles, so they concur. ■

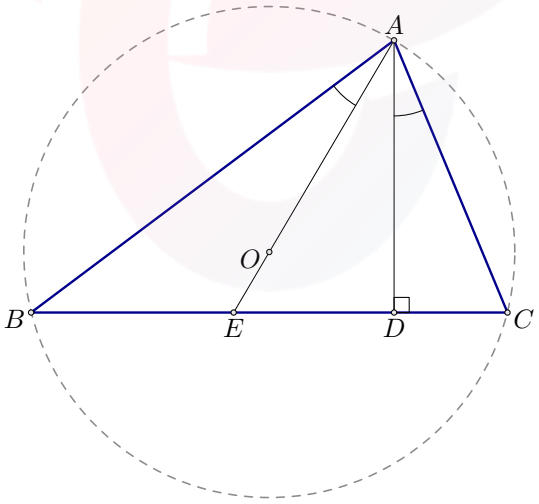
So O and H are isogonal conjugates, the incentre is its own conjugate, and the conjugate of the centroid is the *symmedian point*.

Example 3.
Triangle ABC has $AB = 20$, $AC = 13$, $BC = 21$, and circumcentre O. Line AO meets BC at E. Find BE.

Solution.
The standard way is coordinates. With $B = (0,0)$, $C = (21,0)$ we find $A = (16,12)$, and $O = (\frac{21}{2}, y)$ with $OB = OA$, which gives $y = \frac{8}{3}$. Line AO reaches $y = 0$ at $x = 16 - \frac{9}{7} \cdot \frac{11}{2} = \frac{125}{14}$.
With isogonal lines we never find O. The foot D of the altitude from A has $BD = 16$ and $DC = 5$ (from the coordinates, or from $AB^2 - BD^2 = AC^2 - DC^2$). Since AD and AE are isogonal and the triangle is acute (as $21^2 < 20^2 + 13^2$), Theorem 5 gives

$$\frac{BE}{EC} = \frac{20^2}{13^2} \cdot \frac{DC}{BD} = \frac{400}{169} \cdot \frac{5}{16} = \frac{125}{169},$$

so $BE = 21 \cdot \frac{125}{294} = \frac{125}{14}$. The second method works equally well when the triangle has no convenient coordinates.

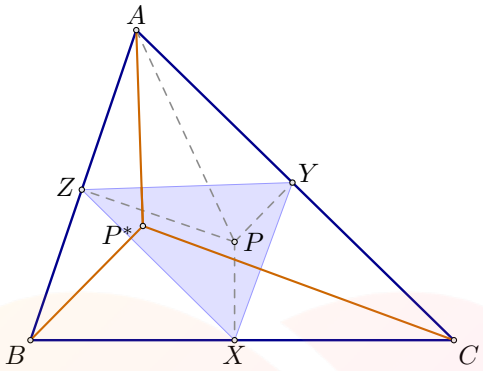


Example 4.
Let P be a point inside an acute triangle ABC, and let X, Y, Z be the feet of the perpendiculars from P to BC, CA, AB. Prove that the line through A perpendicular to YZ, the line through B perpendicular to ZX and the line through C perpendicular to XY are concurrent.

Solution.
Three perpendiculars to the sides of a triangle other than ABC do not give a usable Ceva configuration directly; we identify the lines instead. Since $\angle AZP = \angle AYP = 90^\circ$, the points A, Z, P, Y lie on the circle with diameter AP. As the triangle is acute, the foot Z lies inside

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segment AB, so Z and P both lie on the same side of line AC as B, and $\angle AZY = \angle APY = 90^\circ - \angle PAY$, which is acute.
Let ℓ_A be the line through A perpendicular to YZ, meeting it at U. In the right-angled triangle AZU, the angle between ℓ_A and AB is $90^\circ - \angle AZY = \angle PAY = \angle PAC$. So ℓ_A is the reflection of AP in the bisector of angle A. In the same way the other two lines are the reflections of BP and CP, and by Theorem 6 they meet at the isogonal conjugate P^*.



Common traps.

- Pairing the angles wrongly in Theorem 3. Write the six angles in cyclic order before substituting.
- Stating the angle found by trig Ceva without Lemma 2. A marker wants to see why there is only one solution.
- Using the unsigned forms for points outside the triangle. If A_1 lies on the same arc BC as A with $AA_1 \parallel BC$, and similarly for B_1, C_1, then $A_1B = AC$, $A_1C = AB$ and so on, the product condition of Corollary 4 holds, and yet the three lines through the vertices parallel to the opposite sides are not concurrent.
- In Theorem 5, check whether D and E lie on the segment or on its extension; Theorem 1 holds in both cases, but the angle equalities need checking.

1.4 How to spot this in a contest

1. A point inside a triangle given by angles at two vertices, with the angle at the third vertex asked for: trig Ceva, then Lemma 2.
2. An angle chase in which the angle sum at a point is automatically 360° and gives nothing.
3. Concurrency of cevians whose feet on the opposite sides are hard to locate, for example because they are defined inside another triangle or on the circumcircle.
4. A ratio BD/DC wanted with the angles at A known, or the reverse.
5. Reflections in angle bisectors, or the pairs altitude and circumcentre, median and tangents: isogonal lines and Theorems 5 and 6.

2 Practice problems

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The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Triangle ABC has $\angle BAC = 90°$, $AB = 5$ and $AC = 12$. The point D on BC satisfies $\angle BAD = 30°$. Find BD/DC.

Problem 2 ⋆⋆ A point P inside triangle ABC satisfies $\angle PBC = 24°$, $\angle PBA = 30°$, $\angle PCB = 36°$ and $\angle PCA = 30°$. Find $\angle PAB$.

Problem 3 ⋆⋆ Triangle ABC has $AB = 6$, $AC = 4$ and $BC = 5$. The median from A meets BC at M, and the A-symmedian (the reflection of line AM in the bisector of angle A) meets BC at K. Find MK.

Problem 4 ⋆⋆⋆ Let $0° < \theta < 30°$. Triangle ABC has $\angle ABC = 3\theta$ and $\angle ACB = 120° - \theta$. The point P inside the triangle satisfies $\angle PBC = \theta$ and $\angle PCB = 30°$. Prove that AP bisects angle BAC.

Problem 5 ⋆⋆⋆ Triangle ABC has all three angles between 45° and 90°. The altitudes are AD, BE, CF, and X, Y, Z are the feet of the perpendiculars from D to EF, from E to FD and from F to DE. Prove that AX, BY, CZ are concurrent.

Problem 6 ⋆⋆⋆⋆ Let ABC be a triangle and let α, β, γ be positive angles with $A + \alpha$, $B + \beta$, $C + \gamma$ all less than 180°, and $\beta + \gamma$, $\gamma + \alpha$, $\alpha + \beta$ all less than 180°. Points X, Y, Z lie on the opposite sides of BC, CA, AB from A, B, C respectively, with

$$\angle XBC = \angle ZBA = \beta, \qquad \angle XCB = \angle YCA = \gamma, \qquad \angle YAC = \angle ZAB = \alpha.$$

Prove that AX, BY, CZ are concurrent.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AHSME 1998 Problem 28** ⋆⋆
   Ratio lemma at A with the angles theta and 2 theta.
2. **AMC 12 2001 Problem 24** ⋆⋆
   Call the unknown angle x and apply the sine rule in triangles ABD and ADC.
3. **AIME I 2012 Problem 12** ⋆⋆
   Use the ratio lemma with the 30 degree angles at C, once for D and once for E.
4. **BMO2 2004/05 Problem 2** ⋆⋆
   Show that DE bisects angle ADC by comparing AE/EC with AD/DC.
5. **AIME 1999 Problem 14** ⋆⋆⋆
   Trig Ceva with three equal angles, or split the area at P.
6. **AIME II 2005 Problem 14** ⋆⋆⋆
   AD and AE are isogonal: Theorem 5.

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7. **BMO1 2024/25 Problem 4** ⋆⋆⋆
   P lies on the median, so the ratio lemma gives a relation between the angles at A.

8. **USAMO 1996 Problem 5** ⋆⋆⋆
   Trig Ceva for the unknown angle at B, then Lemma 2.

9. **AIME II 2010 Problem 15** ⋆⋆⋆⋆
   Find the ratio of the sines of the angles BAQ and QAC from triangles at P, then apply the ratio lemma.

10. **BMO2 2016/17 Problem 3** ⋆⋆⋆⋆
    Angle bisector theorem twice; compare the two ratios with the sine rule.

11. **USAMO 2008 Problem 2** ⋆⋆⋆⋆
    Compute the angles ABF and ACF, then identify line AF with trig Ceva or Theorem 5.

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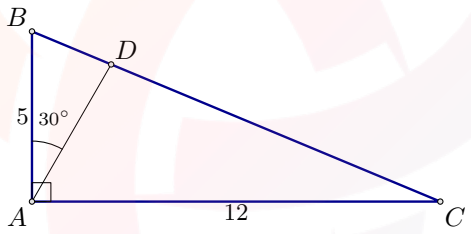
4 Hints

1. Theorem 1 with the angles at A.
2. Write down trig Ceva with $x = \angle PAB$ and look for cancellation. Then use Lemma 2.
3. Theorem 5 with $D = M$.
4. Find all six angles at B and C, then simplify $\sin\theta\cos\theta$.
5. Copy Example 2, with the orthic triangle in place of the contact triangle. You need the angles and the sides of triangle DEF.
6. Let AX meet BC at D and compute BD/DC as a ratio of the areas of ABX and ACX.

5 Solutions

Problem 1

Triangle ABC has $\angle BAC = 90°$, $AB = 5$ and $AC = 12$. The point D on BC satisfies $\angle BAD = 30°$. Find BD/DC.



Here $\angle DAC = 90° - 30° = 60°$. By the ratio lemma,

$$\frac{BD}{DC} = \frac{AB\sin 30°}{AC\sin 60°} = \frac{5\cdot\frac{1}{2}}{12\cdot\frac{\sqrt{3}}{2}} = \frac{5}{12\sqrt{3}} = \frac{5\sqrt{3}}{36}.$$

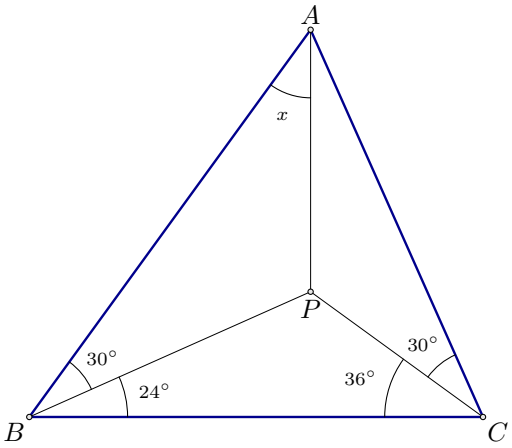
The hypotenuse $BC = 13$ was not needed.

Answer: $\dfrac{5\sqrt{3}}{36}$

Problem 2

A point P inside triangle ABC satisfies $\angle PBC = 24°$, $\angle PBA = 30°$, $\angle PCB = 36°$ and $\angle PCA = 30°$. Find $\angle PAB$.

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We have $B = 54^\circ$, $C = 66^\circ$ and $A = 60^\circ$. Let $x = \angle PAB$, so $\angle PAC = 60^\circ - x$. Trig Ceva gives

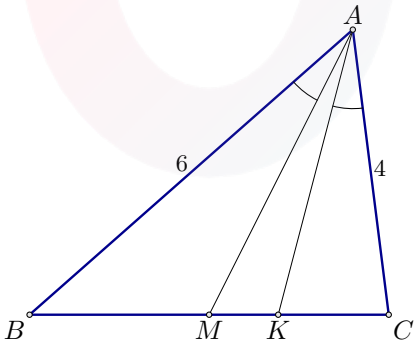
$$\frac{\sin x}{\sin(60^\circ - x)} \cdot \frac{\sin 24^\circ}{\sin 30^\circ} \cdot \frac{\sin 30^\circ}{\sin 36^\circ} = 1, \quad \text{so} \quad \frac{\sin x}{\sin(60^\circ - x)} = \frac{\sin 36^\circ}{\sin 24^\circ}.$$

The value $x = 36^\circ$ satisfies this, since then $60^\circ - x = 24^\circ$. By Lemma 2 with $\theta = 60^\circ$ it is the only solution in $(0^\circ, 60^\circ)$, so $\angle PAB = 36^\circ$.

Answer: 36°

Problem 3

Triangle ABC has $AB = 6$, $AC = 4$ and $BC = 5$. The median from A meets BC at M, and the A-symmedian (the reflection of line AM in the bisector of angle A) meets BC at K. Find MK.



The symmedian meets segment BC, since reflection in the bisector keeps a line inside angle A. By Theorem 5 with $BM/MC = 1$,

$$\frac{BK}{KC} = \frac{AB^2}{AC^2} = \frac{36}{16} = \frac{9}{4},$$

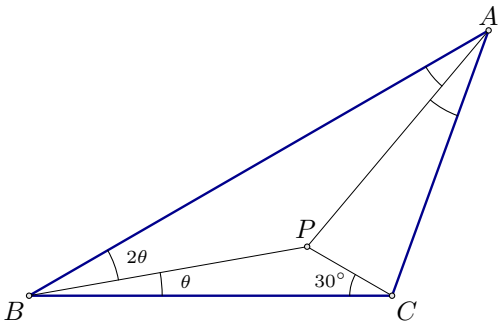
so $BK = 5 \cdot \frac{9}{13} = \frac{45}{13}$. As $BM = \frac{5}{2}$ and $\frac{45}{13} > \frac{5}{2}$, the point K lies between M and C, and $MK = \frac{45}{13} - \frac{5}{2} = \frac{90-65}{26} = \frac{25}{26}$.

Answer: $\dfrac{25}{26}$

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Problem 4

Let $0^\circ < \theta < 30^\circ$. Triangle ABC has $\angle ABC = 3\theta$ and $\angle ACB = 120^\circ - \theta$. The point P inside the triangle satisfies $\angle PBC = \theta$ and $\angle PCB = 30^\circ$. Prove that AP bisects angle BAC.



The angles of the triangle are $B = 3\theta$, $C = 120^\circ - \theta$ and $A = 60^\circ - 2\theta$, all positive for $0^\circ < \theta < 30^\circ$. The point P exists and is inside the triangle, since $\theta < B$ and $30^\circ < C$. We have $\angle PBA = 3\theta - \theta = 2\theta$ and $\angle PCA = 120^\circ - \theta - 30^\circ = 90^\circ - \theta$. Let $x = \angle PAB$, so $\angle PAC = A - x$ with $0^\circ < x < A$.

By Theorem 3,

$$\frac{\sin x}{\sin(A - x)} \cdot \frac{\sin\theta}{\sin 2\theta} \cdot \frac{\sin(90^\circ - \theta)}{\sin 30^\circ} = 1.$$

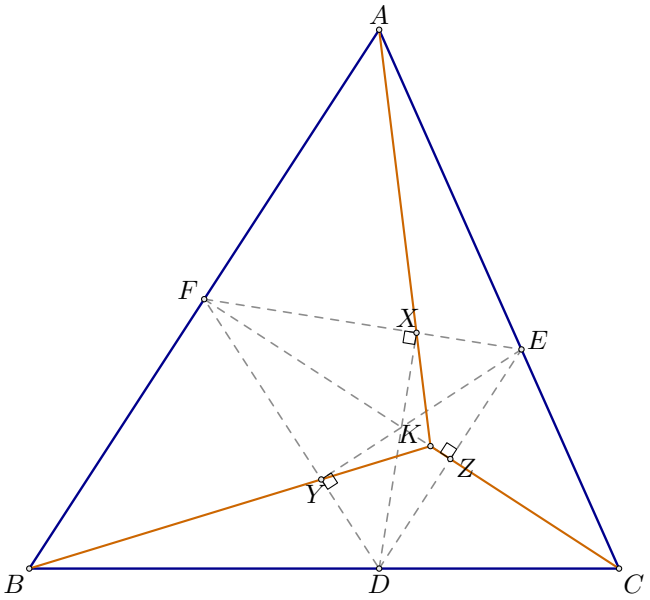
Now $\sin\theta \sin(90^\circ - \theta) = \sin\theta\cos\theta = \frac{1}{2}\sin 2\theta = \sin 30^\circ \sin 2\theta$, so the product of the last two fractions is 1. Hence $\dfrac{\sin x}{\sin(A - x)} = 1 = \dfrac{\sin\frac{A}{2}}{\sin\frac{A}{2}}$, and by Lemma 2 (with θ there equal to A) we get $x = \frac{A}{2}$. So AP bisects angle BAC.

The angle chase alone gives $\angle BPC = 150^\circ - \theta$ and an equation in x that holds for every x, so some trigonometric or auxiliary-point argument is needed.

Problem 5

Triangle ABC has all three angles between 45° and 90°. The altitudes are AD, BE, CF, and X, Y, Z are the feet of the perpendiculars from D to EF, from E to FD and from F to DE. Prove that AX, BY, CZ are concurrent.

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We use the standard facts about the orthic triangle of an acute triangle: triangles AEF, DBF, DEC are similar to ABC, with $AE = AB\cos A$ and $AF = AC\cos A$; the sides of DEF are $EF = a\cos A$, $FD = b\cos B$, $DE = c\cos C$; and its angles are $\angle EDF = 180^\circ - 2A$, $\angle DEF = 180^\circ - 2B$, $\angle DFE = 180^\circ - 2C$. (For instance $\angle AEF = B$ and $\angle CED = B$, so $\angle DEF = 180^\circ - 2B$.)

All angles of ABC exceed 45°, so all angles of DEF are acute, and X lies inside segment EF, hence inside triangle AEF. Then

$$\frac{EX}{XF} = \frac{DE\cos\angle DEF}{DF\cos\angle DFE} = \frac{c\cos C\cdot(-\cos 2B)}{b\cos B\cdot(-\cos 2C)} = \frac{c\cos C\cos 2B}{b\cos B\cos 2C}.$$

By the ratio lemma in triangle AEF,

$$\frac{EX}{XF} = \frac{AE\sin\angle EAX}{AF\sin\angle XAF} = \frac{c\,\sin\angle XAC}{b\,\sin\angle BAX}.$$

Comparing, $\dfrac{\sin\angle BAX}{\sin\angle XAC} = \dfrac{\cos B\cos 2C}{\cos C\cos 2B}$. In the same way $\dfrac{\sin\angle CBY}{\sin\angle YBA} = \dfrac{\cos C\cos 2A}{\cos A\cos 2C}$ and $\dfrac{\sin\angle ACZ}{\sin\angle ZCB} = \dfrac{\cos A\cos 2B}{\cos B\cos 2A}$.
The product of the three fractions is 1. Each of AX, BY, CZ meets the opposite side at an interior point, since X, Y, Z are inside the triangle, so by Theorem 3 the three lines are concurrent.

The hypothesis on the angles is used once, to put X, Y, Z inside the sides of DEF. Without it one of the feet lies outside, and a signed version of the argument is needed.

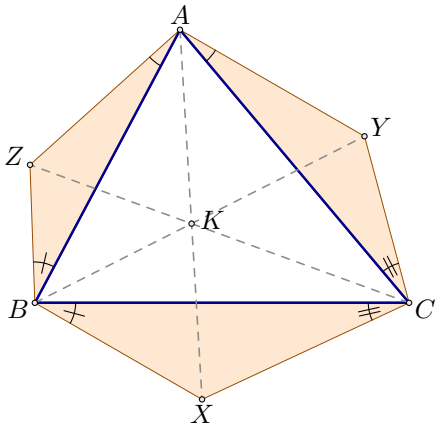
Problem 6

Let ABC be a triangle and let α, β, γ be positive angles with $A+\alpha$, $B+\beta$, $C+\gamma$ all less than 180°, and $\beta+\gamma$, $\gamma+\alpha$, $\alpha+\beta$ all less than 180°. Points X, Y, Z lie on the opposite sides of BC, CA, AB from A, B, C respectively, with

$$\angle XBC = \angle ZBA = \beta, \qquad \angle XCB = \angle YCA = \gamma, \qquad \angle YAC = \angle ZAB = \alpha.$$

Prove that AX, BY, CZ are concurrent.

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The conditions $\beta+\gamma<180^\circ$ etc. make the triangles XBC, YCA, ZAB exist. Since X is on the other side of BC from A, and $\angle ABX = B+\beta$ and $\angle ACX = C+\gamma$ are less than 180°, the quadrilateral $ABXC$ is convex, so its diagonals AX and BC meet at a point D inside both. Similarly BY meets CA at an interior point E and CZ meets AB at an interior point F.

The distances from B and C to line AX are in the ratio $BD:DC$, so

$$\frac{BD}{DC}=\frac{[ABX]}{[ACX]}=\frac{\frac12 AB\cdot BX\sin(B+\beta)}{\frac12 AC\cdot CX\sin(C+\gamma)}.$$

The sine rule in triangle XBC gives $BX/CX=\sin\gamma/\sin\beta$. Hence, writing a,b,c for the sides,

$$\frac{BD}{DC}=\frac{c\,\sin\gamma\,\sin(B+\beta)}{b\,\sin\beta\,\sin(C+\gamma)}.$$

The same computation at the other vertices gives

$$\frac{CE}{EA}=\frac{a\,\sin\alpha\,\sin(C+\gamma)}{c\,\sin\gamma\,\sin(A+\alpha)},\qquad \frac{AF}{FB}=\frac{b\,\sin\beta\,\sin(A+\alpha)}{a\,\sin\alpha\,\sin(B+\beta)}.$$

Every factor cancels in the product, so $\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}=1$, and since D, E, F are interior points of the sides, Ceva's theorem shows that AX, BY, CZ are concurrent. ■

This is Jacobi's theorem. With $\alpha=\beta=\gamma$ the triangles XBC, YCA, ZAB are isosceles and similar (Kiepert's theorem); the value 30° gives the lines from the vertices to the centres of the outer equilateral triangles.